YILI YANG, PhD

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Applied AI Researcher | Deep Learning, Time Series, Production ML

SKILLS

DOMAINS

Machine Learning & Deep Learning|Time-series modelling
Multimodal Learning|Image Segmentation
Object Detection|Earth Foundation Models
AI for Science|Scientific Data & Image Processing

TECH STACK

PyTorch|Scikit-learn
Google Cloud Platform|Google Earth Engine

ENGINEERING

Docker|GitHub Actions
Git|GitHub
Agentic Coding|Harness Engineering

EDUCATION

UNIVERSITY OF EDINBURGH

PhD Petrophysics,2021
MSc Geology, 2016
BSc Geoscience, 2015

STATS

PUBLICATIONS

Journal Papers: 13
Conferences: 8
Data Sets: 1

ACADEMIC SERVICES

Invited Talks & Panels: 17
Peer-reviews: 8
Mentees: 9

LINKS

Personal Website
LinkedIn || ORCID
Google Scholar || GitHub

EXPERIENCE

DATA SCIENTIST | Woodwell Climate Research Center

Jan 2022 - Present | Falmouth, MA, USA

- **Technical lead**: Led a continental-scale deep-learning programme end to end, from research design through to published results
- **Production ML**: Optimised deep-learning inference pipelines over very large datasets on Google Cloud Platform
- **Time series**: Time-series imputation and multi-domain prediction with machine learning
- **Frameworks**: Designed and built a benchmark training dataset now used as a shared resource for deep-learning research
- **Foundation models**: Built a zero-shot interactive segmentation workflow on the Segment Anything Model
- **Leadership**: Ran concurrent modelling collaborations with three university research groups; lead the annual machine-learning workshop series

DATA SCIENCE FELLOW | Faculty.AI

Sep 2021 - Dec 2021 | London, UK

- Commercial data-science fellowship in machine learning and AI, including an industry client placement

DOCTORAL RESEARCHER |ICCR

International Centre for Carbonate Reservoirs
Sep 2016 - Apr 2021 | Edinburgh, UK

- PhD developing novel computational methods for multiphase flow in porous media
- Deep-learning image segmentation applied to three-dimensional imaging data

SELECTED PUBLICATIONS

- Segment anything model can not segment anything: Assessing ai foundation model's generalizability in permafrost mapping — *Remote Sensing*, 2024.
- Data-driven modeling of carbon dioxide and methane fluxes across the Arctic-boreal region: recent achievements and future opportunities — *submitted to Global Change Biology*.
- A collaborative and scalable geospatial data set for arctic retrogressive thaw slumps with data standards — *Scientific Data*, 2025.

AWARDS & GRANTS

FUNDING FOR CLIMATE SOLUTIONS

Funding for Climate Solutions, Woodwell Climate Research Center
$99,749| 2025-2026 | A Generic Climate AI Framework for Multi-domain Time Series Prediction

MULTIDISCIPLINARY COLLABORATIONS

GOOGLE × WOODWELL

Understanding Permafrosts, AlphaEarth and Satellite Embedding